

Safely Expanding the Verification Frontier of Scientific Agents

安全扩展科学 Agent 的验证边界

GOAI 2026 AI for Research — Open Exploration

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1. Research problem and proposed contribution

Scientific agents must distinguish a candidate established by an executable check, a candidate refuted by a check, and a candidate outside current verifier coverage. Treating the third case as “no error found” creates a silent pass; merely abstaining leaves the workflow stuck.

Can model-proposed, human-approved verifier extensions increase the safe executable coverage of a frozen scientific verification stack on sealed transfer tasks?

Contest-period proposed contribution

- Typed coverage gaps and fail-closed three-state outcomes;
- bounded specification or verifier proposals;
- owner and independent expert-AI approval;
- positive, negative, fail-closed, and hidden-transfer tests;
- coverage-delta, regression, activation, and rollback receipts.

Novelty boundary: this project does not claim to invent dimensional analysis, SymPy, Lean, RLVR, agent abstention, or the verification frontier. The deliverable is the environment and verification signal, not scientific discovery, recursive self-improvement, model capability, or AGI.

2. Exploration environment and human gate

Outcome	Contract
accepted	An applicable deterministic check establishes the candidate.
rejected	An applicable check establishes a concrete failure.
abstain	No applicable executable check can decide; never a silent pass.

observe → propose → verify
accepted: keep | rejected: revise or stop | abstain: classify the gap and propose a bounded extension
→ owner + expert-AI gate → executable tests → re-verify → activate locally or preserve abstention

Fail-closed invariants

- A model cannot approve its own extension;
- a reviewer who saw aggregate confirmatory outcomes does not count;
- missing approval or failed tests preserve abstention;
- open-problem controls can never be promoted;
- every terminal receipt preserves candidateOnly:true and canClaimAGI:false, with winner gates false.

Three deterministic domains

Domain	Coverage contract
Physics	SI, affine units, vectors, uncertainty, conservation, frames
Symbolic	Domains, assumptions, singularities, piecewise bounds, inequalities
Lean	Pinned Lean 4.24.0 + Mathlib kernel and claim/formal alignment

Content-addressed evidence DAG

proposal# → owner-review# + expert-AI-review# → typed-tests# → activation#
→ transfer# → protected-suite# → rollback# → extension-chain#

The public no-login demo exposes public verifiers, reference episodes, synthetic gate mechanics, and seal metadata—never sealed tasks, golds, or credentials.

3. Preregistered benchmark and current milestones

Component	Physics	Symbolic	Lean	Total
Frontier valid+safety pairs	20	20	20	60
Auxiliary transfer valid+safety pairs	20	20	20	60
Already-covered control pairs	4	4	4	12
Full study tasks	88	88	88	264

Primary endpoint: Safe Frontier-Pair Accuracy (SFPA)

A pair succeeds only when its valid member is correctly machine-decided, its safety sibling is rejected or abstained, and the responsible extension passes transfer and regression gates. SFPA covers 60 pairs, but inference clusters over at least 30 independent extension/generator families; parameter variants and sampling replicates do not inflate independent N.

Preregistered winner-level gates

- Zero unsafe acceptance among 60 frontier safety siblings;
- delta_SFPA at least +20 percentage points over the strongest non-oracle baseline;
- 95% cluster-bootstrap CI lower bound > 0 and paired sign-flip $p < 0.05$;
- positive point estimate in each required model family versus its own strongest baseline;
- every counted extension passes two sealed valid transfers and paired safety tasks;
- no covered-control or protected-suite regression.

Instrument evidence completed by August 1, 2026

Milestone	Status
Public development/regression pack	150 rows; 50 per domain
Stage A + guarded Pro6000 lane	24 families (8/domain); 30 tasks bound once; 3 open controls non-promotable; no model outcome/approval yet
Real Lean development validation	150/150
Content-addressed receipt protocol	3 chains / 34 receipts / 60 blobs; benchmark 7/7
CPU-only Protocol Twin	B0-B6; 8 groups / 13 variants; 2,160 cells; 0 model/network calls
Study Root v3	756 arm / 1,404 ablation / 6 transfer descendants; 24 serialization variants + 164 invalid mutations; 24,000 sims
Synthetic rehearsal	144 tasks / 72 pairs; confirmatoryEligible:false
Rehearsal validation	144/144; 48 Lean items; only 15 leaked families

Falsify the winner claim if: any unsafe acceptance occurs; any required model family loses to its strongest baseline; any receipt link or covered control regresses; or the CI / p-value gates fail.

These establish instrument integrity, not frontier-expansion efficacy.

4. Baselines, related work, disclosure, and continuation

Required experimental arms

Raw model; fixed three-state verifier; fixed verifier plus equal-budget refinement; act-or-abstain without executable extension; budget-matched human-only extension; proposed human-gated system; expert-authored oracle ceiling. Key ablations remove the human gate, executable patch, transfer obligation, explicit abstention, and each verifier tier. The CPU-only twin checks B1/B5 replay, B2 revised input, an A6 sentinel, cell completeness, and fixture budget fields. It is not bound to a scored study or an efficacy experiment.

Related work and novelty risk

Direct work includes RLVP (arXiv:2607.10474), EG-VAR (arXiv:2607.12650), Recursive Epistemic Engines and its Novelty Horizon, and AgentAbstain (arXiv:2607.10059). The largest risk is prior guarded verifier-expansion work. The defensible contribution is prospective empirical evidence and reproducible instrumentation—not invention of the frontier.

Disclosure

Pre-existing: Sophia SI/symbolic verifiers, Lean checking/evaluations, provenance and no-overclaim gates, and prior candidate ladder evidence.

Contest-period: typed frontier protocol, matched-pair generator and seal, content-addressed receipt protocol, human-gate contracts, the 24-family Stage A programme, guarded local-model runner, statistics, bilingual materials, and hosted-demo package.

Continuation condition

Current recommendation: NO-GO for confirmatory execution; CONDITIONAL GO for preliminary upload as an infrastructure proposal. Scoring starts only after a genuinely private 30-family benchmark, prompts, models, budgets, and extension bundle are frozen, and the receipt protocol is instantiated with real independent reviews, hidden transfer, protected-suite, and rollback runs.

```
./run_all.sh  
python3 v2/run_model_attempts.py --dry-run
```

Limits: no confirmatory benchmark or efficacy result exists; the rehearsal is constructed and structurally leaked; Stage A local proposals, dual reviews, and executable extension tests are incomplete; human and domain-expert review remain necessary. candidateOnly:true; canClaimAGI:false; winnerLevelEligible:false; winnerLevelGateMet:false.